

MTH209: Assignment 6

t-SNE and distance

In your .zip folders there is a dataset “assignment_dataset.csv” (remember to extract the files before you can load this dataset). This contains 6 columns:

- the first five columns are numerical features
- the 6th column is a categorical variable: **type**.

Write a code to fit t-SNE with perplexity = 50 and dimension = 2. Then, return a vector of length two where:

1. the first element is the Euclidean distance between the 15th and 20th observation in the embedding space.
2. the second element is the minimum Euclidean distance of the 15th observation from any observation of type = 2, in the embedding space.

This vector of length two, must be called **answer**.

```
data <- read.csv("assignment_dataset.csv")

# do tsne with perplexity = 50 and dimension = 2

# find distance between 15th and 20th
dist1 <- ...

# find minimum distance between 15th and type 2 observations
dist2 <- ...

## this must be the last line of the code
answer <- c(dist1, dist2)
```

INSTRUCTIONS: Upload the file with the name **assignment6.R** on helloIITK. Make sure your code:

1. does NOT contain a **set.seed** command
2. does NOT print anything